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ANTICLOCKWISE ROTATION OF ITALY BETWEEN THE EOCENE AND MIOCENE: PALAEOMAGNETIC EVIDENCE FROM THE COLLI EUGANEI, ITALY

H. SOFFEL
Institut für Angewandte Geophysik, Universität München, Germany

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The palaeomagnetism of 230 cores from 25 sites from the Colli Euganei in Northern Italy indicates that between the Eocene and Miocene Italy rotated anticlockwise through an angle of 55°. This is in agreement with the palaeomagnetic investigations by various researchers in the Netherlands on late Palaezoic and Mesozoic rocks from Northern Italy.

1. Introduction

Volcanism in the Colli Euganei in Northern Italy, south of Padova, is characterized by two eruptive cycles. The first cycle was dated as Lutetian and Priabonian, which lies in the middle and upper Eocene (50–45 m.a.) according to stratigraphic methods [1]. For specimens mainly of the second cycle radiometric age determinations (K/Ar) by Borsi et al. [2] are available yielding a middle to upper Oligocene age (32 m.a.). This age has been confirmed for one volcanic unit of the second cycle by stratigraphic investigations of Dieni and Decima [1]. In the earlier palaeomagnetic work by De Boer and Schouten [3] samples of the second eruptive cycle were presumed to have an age between upper Miocene and upper Pliocene, which is 15–5 m.a. All these age determinations suggest an age of 50–45 m.a. for the first eruptive cycle and an age of roughly 33–10 m.a. for the second and more active eruptive cycle, which might have reached its climax in lower Miocene (25 m.a.). Therefore a hiatus of the order of about 20 m.a. can be expected between the maxima of the two eruptive cycles in the Colli Euganei.

2. Sampling

The sampling area is shown in fig. 1 together with the localities of the sampling sites. The numbers refer to the site numbers used later in this paper. 25 localities were sampled which are spread equally over the whole volcanic area yielding 230 cores and 600 specimens for palaeomagnetic measurements. The mean geographic coordinates of the sampling area are: $\lambda = 11.7°E$, $\varphi = 45.3°N$.

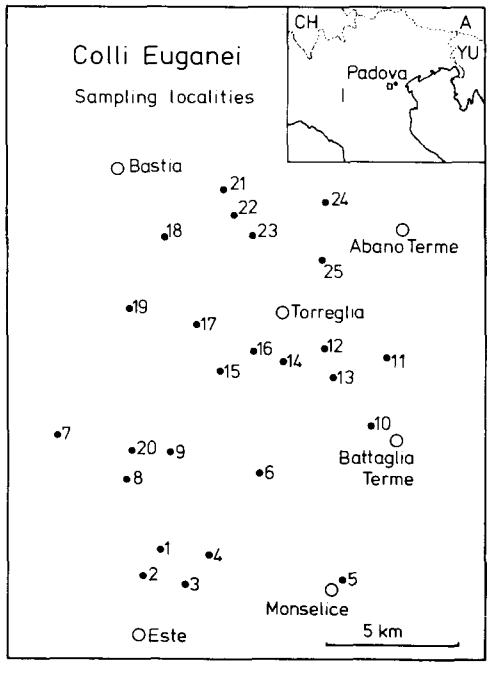


Fig. 1. Sampling area and sampling sites in the Colli Euganei.

3. Palaeomagnetism

Only a brief review of the palaeomagnetic results which are of interest for the postulated rotation of Italy are presented here. A more detailed paper including the petrography of the rocks and some related rockmagnetic properties will be published later elsewhere.

Fig. 2 and table 1 show the mean directions of remanent magnetization after selective AC-demagnetization in fields up to 1200 Oe for the different sites. Numbers refer to the sites shown in fig. 1. Site numbers 1, 2 and 14 were omitted from the final conclusions because of their unstable magnetization which was evident from their behaviour during AC cleaning and the large within-site scatter which they display. Almost all sites show reversed polarity. Three sites with normal polarity (3, 15 and 18) were projected on the lower hemisphere in fig. 2 and symbolized with a prime. Samples of site number 15 showed both polarities although no petrological or other difference has so far been detected.

The distribution of the site mean directions is a non-Fisherian with inclinations around -40° and declinations ranging from 225° to 105°, the scatter of the declinations being much larger than that of the inclinations. The mean of all sites (excluding 1, 2 and 14 for reasons mentioned previously) is $D = 172.4^\circ$, $I = -47.1^\circ$, $K = 7.8$ and $\alpha_{95} = 10.5^\circ$, which is not plotted in fig. 2.

Fig. 3 contains pole positions on the Northern hemisphere from lower Tertiary (small squares) and upper Tertiary (small circles) according to a compilation of Tertiary data for stable Europe by Duncan et al. [4]. (For the numbers of poles, see the same paper.) The mean direction ($D = 172.4^\circ$, $I = -47.1^\circ$) of the non-Fisherian distribution of site means of the Colli Euganei yields a palaeomagnetic pole (large square in fig. 3) which is distant from all other Tertiary palaeomagnetic pole positions for stable Europe.

However, two individual Fisherian distributions can be obtained from the data plotted in fig. 2 if a separation is made between sites having declinations larger or smaller than 150° respectively. The precision parameter K for the two individual groups is hereby increased with respect to K of the non-Fisherian distribution. The two mean directions which can be derived for the individual groups are: $D = 191.6^\circ$, $I = -48.2^\circ$,

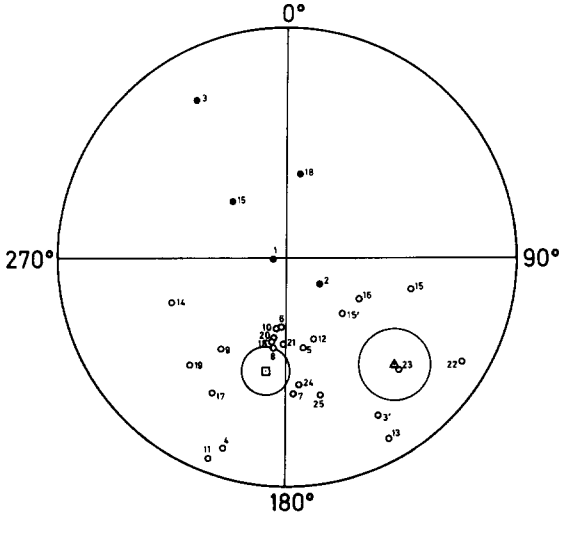


Fig. 2. Equal area projection of site mean directions of remanent magnetization after AC cleaning. Numbers refer to the site numbers of fig. 1. Open symbols: negative I; closed symbols: positive I. Square: mean of sites of group I (D larger than 150°). Triangle: mean of sites of group II (D smaller than 150°). Sites with a prime are original directions of normal polarity projected on the other hemisphere.

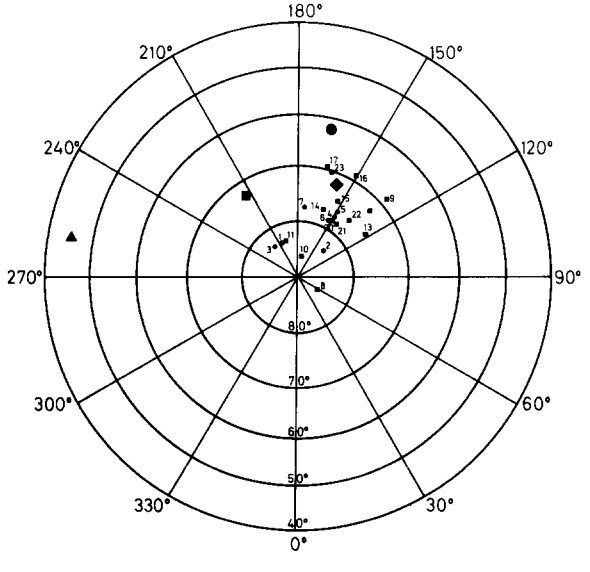


Fig. 3. Pole positions for stable Europe compiled by [4]. Small squares: lower Tertiary, small circles: upper Tertiary. Large circle: Cretaceous pole position according to [7]. Large square: mean pole position of all sites of the Colli Euganei. Diamond: pole position of sites of group I. Triangle: pole position of sites of group II.

Table 1
Mean directions of remanence of the different sites after AC demagnetization.

Site no.	Group	N	No.	D	I	K	α_{95}	RM(γ)	H_{ent}(Oe)
1	–	10	27	269.7	+85.2	2.5	17.0	1	800
2	–	10	32	121.5	+75.2	4.0	12.4	1	700
3	II	9	17	331.8	+21.2	26.8	6.6	10	50
4	I	9	25	199.5	−13.2	18.1	6.6	5	400
5	I	10	29	171.4	−57.0	11.6	7.6	6	400–800
6	I	7	18	184.3	−64.4	26.2	6.4	2	400–800
7	I	10	24	178.5	−40.4	42.6	4.4	12	200–600
8	I	9	26	189.3	−57.6	54.4	3.7	30	300–500
9	I	8	18	217.6	−49.5	11.7	9.6	1	800
10	I	8	12	188.7	−65.5	175.1	3.1	150	400
11	I	11	30	202.6	− 6.5	22.1	5.4	5	200–500
12	I	13	34	163.4	−59.1	74.7	2.8	50	300–800
13	II	9	12	151.7	−10.5	36.1	6.7	60	200–1000
14	–	11	12	250.1	−45.7	2.6	24.9	1	200–600
15 n	II	4	11	318.5	+62.6	124.7	3.8	2	200–1000
15 r	II	3	8	206.1	−44.5	16.6	12.2	2	200–1000
16	II	12	28	121.2	−60.7	26.9	5.1	15	200–600
17	I	10	17	209.9	−33.2	4.8	15.4	0.5	500–900
18	I	11	28	10.7	+59.1	6.9	10.1	1	200–400
19	I	6	14	225.0	−37.0	4.2	18.2	1	50–200
20	I	8	16	189.4	−61.5	44.6	5.2	13	200–400
21	I	10	26	183.5	−58.8	20.8	6.0	20	200–500
22	II	10	16	122.3	−13.3	46.0	5.2	10	50–100
23	II	9	16	137.1	−31.8	7.9	12.5	2	200–700
24	I	10	26	175.7	−43.8	38.3	4.4	8	300–600
25	I	2	2	168.1	−38.4	20.8	21.7	2	100

N: number of cores; No.: number of specimens; D: declination (North to East); I: inclination; K: precision parameter; α_{95}: radius of circle of confidence; RM(γ): remanent magnetization after AC demagnetization in γ. H_{ent}: peak values of the demagnetizing fields.

K = 14.6, α_{95} = 9.2° for group I and D = 135.1°, I = −33.7°, K = 12.1, α_{95} = 15.2° for group II. They are plotted in fig. 2. The pole position of group I is plotted in fig. 3 as the large diamond. It fits well with other data from lower to middle Tertiary for stable Europe essentially confirming the results of De Boer and Schouten [2] and the other age determinations mentioned previously. The pole positions of group II (the large triangle in fig. 3) is far from the position of other Tertiary poles for stable Europe, partly with regard to its latitude but more with regard to its longitude. The deflection will be discussed below in connection with the rotation of Italy.

4. Rotation of Italy and discussion of the Colli Euganei results

An anticlockwise rotation of Italy of about 50° since the Permian [5] and, from more detailed studies of other authors (see e.g. [6]) since the Triassic, has been established from palaeomagnetic investigations on rocks from Northern Italy, mainly by workers in the Netherlands. However, palaeomagnetic data of rocks from Southern Italy are not yet available. Fig. 4 shows the expected mean directions of remanence (only reversed directions) for the Northern Italy area derived from palaeomagnetic pole positions of stable Europe. Square with vertical lines: mean direction for the Permian of stable Europe according to a compilation by [6]; square with horizontal lines: Cretaceous direction derived from data

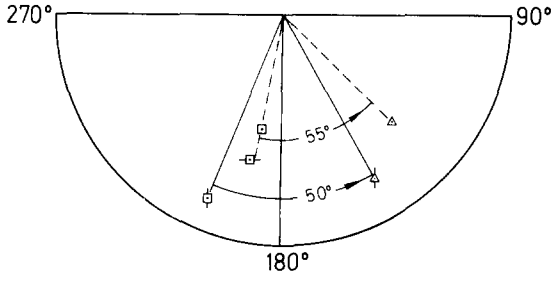


Fig. 4. Mean directions of remanence (negative *I*) for Northern Italy. Square and triangle: mean directions of group I and group II respectively of the Colli Euganei sites. Square with horizontal lines: Cretaceous direction according to [7]. Square with vertical lines: Permian direction for stable Europe compiled by [6]. Triangle with vertical lines: direction of the Permian Bolzano Rhyolite after [6]. Arrows indicate the amount and direction of rotation of Italy.

by Kramov [7]; square: Colli Euganei, mean direction of group I; triangle with vertical lines: direction derived from new data by Zijderveld et al. [6] for the Permian Bolzano Rhyolites; triangle: Colli Euganei, mean direction of group II. The deviation of the mean direction of the Bolzano Rhyolites from the direction derived from the Permian pole positions for stable Europe led the previously cited authors ([5, 6] and others) to the assumption of an anticlockwise rotation of Italy of about 50° since Permian times. The mean direction of the Colli Euganei sites of group II can be brought into coincidence with Tertiary (square) and Cretaceous directions (square with horizontal line) by an anticlockwise rotation of roughly the same amount (55°). Considering the time span discussed earlier for the volcanism of the Colli Euganei and its separation into two main periods of activity it is suggested that the rotation of Italy took place between Eocene and Miocene. This conclusion implies that sites from group II belong to the first eruptive cycle of the Colli Euganei. This is supported by only one radiometric age by Borsi et al. [2] giving an age of 42 ± 1.5 m.a. for samples taken at locality number 16 which belongs to group II. The younger radiometric ages all belong to sites of group I, which are therefore associated with the second and younger cycle of volcanic activity in the Colli Euganei.

Whether the rotation of Italy in the Tertiary has also been accompanied by a movement to the north with respect to stable Europe cannot be deduced from the Colli Euganei data because of their scatter. However, there is some indication for such a process as the mean inclinations are slightly too low compared with those of stable Europe.

The mean drift rate computed for the southern end of Italy during a rotation of around 50° would be about 800 km in roughly 20 m.a. which is 4 cm/yr. This value is in agreement with those deduced by various authors for similar plate movements.

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